Mathematics Stage 3 -- Year B -- Unit 31 Number & Multiplicative Relations

KLA: Mathematics | Stage 3 | Year 6 | Term: _____ | Duration: 2-3 weeks | Teacher: _____ | Year: _____

Syllabus outcomes

Code	Outcome descriptor
MAO-WM-01	Develops understanding and fluency through exploring, applying and communicating mathematical reasoning.
MA3-RN-01	Applies place value to represent natural numbers to billions.
MA3-RN-02	Compares and orders decimals up to 3 decimal places.
MA3-RN-03	Determines percentages of quantities and benchmark fraction/decimal/percentage equivalents.
MA3-MR-01	Selects and applies strategies to solve multiplication and division problems.
MA3-MR-02	Constructs and completes number sentences, applying the order of operations.

Syllabus content points (addressed through SOLO tracker)

All content points below are covered by students working independently through the SOLO tracker. Mini masterclasses target specific outcomes based on live tracker data.

Red -- Foundation	Yellow -- On-level	Green -- Extension
Numbers A & Multiplicative A (place value, decimals, ×/÷) Partition, regroup and rename whole numbers to 1 billion in standard and non-standard forms. Express and interpret decimals to thousandths; name place value to 3 dp; interpret trailing zeros. Multiply 2- and 3-digit numbers by 2-digit numbers using the area model and algorithm. Divide a 3+ digit number by a one-digit divisor, interpreting remainders.	**Numbers B & Multiplicative B (percentages, patterns, sentences)** Percentages: % and 100% as the whole; represent common percentages as fractions and decimals. Find 10% of a quantity (10% is one-tenth of 100%). Build a table of values from a geometric pattern and describe it as multiplication. Determine the rule relating the two rows of a table (algebraic reasoning). Multiply and divide decimals by powers of 10. Complete number sentences with missing numbers using inverse operations.	**Extended -- Stage 4 Number content** Find a percentage of a quantity and calculate percentage increase and decrease (MA4-FRC-C-01). Apply the order of operations with grouping symbols to the four operations (MA4-INT-C-01). Solve simple (up to 2-step) linear equations using pronumerals and inverse operations (MA4-EQU-C-01).

Teaching model

Students work independently through the Word Labs SOLO Tracker (wordlabs.app/solo) across differentiated bands -- Grow (resources), Know (practice), Show (auto-marked assessments). The teacher monitors live dashboard data and pulls small groups for targeted mini masterclasses (15-40 min) based on which outcomes have the most incomplete Show results. Remaining session time is used for incidental one-on-one and small group teaching in response to student questions and tracker data.

Working Mathematically (MAO-WM-01)

Process	How it appears in this unit
Communicating	Students describe numbers using place value to billions and thousandths, and describe percentages and pattern rules using multiplication language.
Understanding and fluency	Students partition numbers, read decimals to thousandths, multiply and divide accurately, and shift digits when multiplying/dividing by powers of 10.
Reasoning	Students reason about place-value relationships, determine the rule of a table, and use inverse operations to justify a missing number.
Problem solving	Students solve multiplication and division word problems, find percentages of quantities, and (Stage 4) apply order of operations and solve simple equations.

Differentiation

Support	HPGE / Extension
Students in this group:	Students in this group:

Pre-test

Date	
Format	
Band placements / key findings	

Post-test / unit evaluation

Date	
Key findings	
Students who need further consolidation	
Next steps	

Resources

See Appendix for Resources

Word Labs SOLO Tracker: wordlabs.app/solo | NSW DoE Stage 3 Year B Unit 31 | NESA Mathematics K-10 Syllabus (2022)

Unit 31 -- Outcome Teaching Record

Find the outcome/s you taught, write your notes, sign and date. Order follows the outcome sequence, not your teaching calendar.

Code	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
R1	Partition, regroup and rename whole numbers to 1 billion.	RN-A Whole numbers: 1000 thousands = 1 million, 1000 millions = 1 billion; regroup; partition to 1 billion in non-standard forms MA3-RN-01 Year A	Mini Lesson 1		
R2	Express and interpret decimals to thousandths (place value to 3 dp; trailing zeros).	RN-A Decimals: express thousandths as decimals; interpret notation; place value to 3 dp; interpret trailing zeros MA3-RN-02 Year A	Mini Lesson 2		
R3	Multiply 2- and 3-digit numbers by 2-digit numbers (area model, algorithm, word problems).	MR-A: factorise; extend area model to 2-digit x 2-digit; multiplication algorithm; multiplication word problems MA3-MR-01 Year A	Mini Lesson 8		
R4	Divide a 3+ digit number by a one-digit divisor, interpreting remainders.	MR-A: division word problems; use and interpret remainders; divide using digital technologies MA3-MR-01 Year A	Mini Lesson 9		
Y1	Percentages: % and 100% as the whole; common percentages as fractions and decimals.	RN-B Percentages: % means percent and 100% is the whole; represent common percentages of quantities/lengths as fractions and decimals MA3-RN-03 Year B	Suggested: pair with Y2 Mini Lesson 3		
Y2	Find 10% of a quantity (10% is one-tenth of 100%).	RN-B Percentages: recognise 10% is one-tenth of 100% and use this to find 10% of a quantity MA3-RN-03 Year B	Suggested: pair with Y1 Mini Lesson 4		
Y3	Build a table of values from a geometric pattern and describe it as multiplication.	MR-B Patterns: use a geometric pattern of multiples to create a table of values; describe in terms of multiplication MA3-MR-01 Year B	Suggested: pair with Y4 Mini Lesson 6		
Y4	Determine the rule relating the two rows of a table (algebraic reasoning).	MR-B Patterns: determine a rule describing the relationship between the bottom and top numbers in a table MA3-MR-01 Year B	Suggested: pair with Y3 Mini Lesson 7		
Y5	Multiply and divide decimals by powers of 10.	MR-B: compare the relative place value of digits to multiply and divide a decimal by powers of 10 MA3-MR-01 Year B	Mini Lesson 11		
Y6	Complete number sentences with missing numbers using inverse operations.	MR-B Number sentences: complete multi-operation sentences by calculating missing numbers; use inverse operations MA3-MR-02 Year B	Mini Lesson 12		
G1	Find a percentage of a quantity; calculate percentage increase and decrease.	Stage 4 FRC-A: one quantity as a percentage of another; percentage increase/decrease; solve percentage problems MA4-FRC-C-01 Stage 4 (group A)	Mini Lesson 5		
G2	Apply the order of operations with grouping symbols to the four operations.	Stage 4 INT-A: apply the 4 operations; grouping symbols; order of operations to evaluate expressions MA4-INT-C-01 Stage 4 (group A)	Mini Lesson 10		

Code	Outcome	Syllabus content point/s addressed	Mini masterclass teaching notes	Sign	Date
G3	Solve simple (up to 2-step) linear equations using pronumerals and inverse operations.	Stage 4 EQU-A: represent unknowns with pronumerals; describe number sentences as equations; solve up to 2-step linear equations; word problems MA4-EQU-C-01 Stage 4 (group A)	**Mini Lesson 13**		

Syllabus outcomes and content from Mathematics K-10 Syllabus (2022) copyright NSW Education Standards Authority (NESA) for and on behalf of the Crown in right of the State of New South Wales, 2024. RN = Represents Numbers (whole numbers, decimals, percentages). MR = Multiplicative Relations (multiplication, division, number sentences). FRC/INT/EQU = Stage 4 Fractions-decimals-percentages / Computation / Equations. A/B = syllabus content groups (A place-value & ×÷ fluency, consolidated; B percentages, pattern rules & number sentences, taught as home content).

Unit 31 -- Mini Lesson Sequence

Number & Multiplicative Relations | Stage 3 Year B

Supplement to the SOLO Teacher Program (pages 1-2). Print pages 1-2 only for annotation. Reference these lesson plans when planning mini masterclasses or when a reviewer requires evidence of planned lesson content.

Red band -- Foundation -- Mini Lessons 1-9	Yellow band -- On-level -- Mini Lessons 3-12	Green band -- Extension -- Mini Lessons 5-13

DoE lesson links: Lessons marked DoE Lesson X draw directly from NSW DoE Mathematics Stage 3 Unit 31. Steps have been condensed to 20-40 minutes for mini masterclass delivery. Resources referenced by their original resource numbers from the DoE unit.

Mini Lesson 1 Partitioning and renaming numbers to 1 billion [HANDS-ON]

R1 | Red band | DoE Lesson 1 (adapted)

Duration: 25-30 min	Outcomes: R1	Syllabus: MA3-RN-01, MAO-WM-01 | RN-A: partition, regroup and rename numbers to 1 billion

Linked to: DoE Lesson 1 (adapted) -- Lesson 1 -- Regrouping and partitioning numbers | Key resources: Place-value chart, number expander, MAB or place-value cards

Physical materials needed: place-value chart, number expander, place-value cards / MAB

Learning intention	Success criteria
Students are learning to: partition, regroup and rename whole numbers to 1 billion in standard and non-standard forms	Students can: * I know 1000 thousands is 1 million and 1000 millions is 1 billion. * I can partition a large number by place value (e.g. 3 450 000 = 3 000 000 + 450 000). * I can rename a number in a non-standard form (e.g. 3 450 000 = 34 hundred-thousands + 5 ten-thousands).

Mini masterclass structure (25-30 min)

Activate (4 min): Build the place-value pattern: ones, thousands, millions, billions. Ask how many thousands make a million.

Model (10 min): Partition 3 405 000 by place value, then regroup it a non-standard way (e.g. 3405 thousands). Read it aloud and write it in expanded form.

Guided practice (10 min): Students partition and rename a set of large numbers in standard and non-standard forms using a place-value chart.

Check (3 min): Exit question: partition 1 250 000 two different ways and state how many thousands are in 5 million.

Key vocabulary	Differentiation
place value, partition, regroup, rename, thousand, million, billion, standard form, non-standard form	**Support:** Use a place-value chart and number expander; start with thousands before extending to millions. **Extension:** Rename numbers in several non-standard forms and justify which is most useful for a calculation.

Assessment	Teaching notes (annotate here)
Students partition, regroup and rename numbers to 1 billion — links to R1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 2 Decimals to thousandths

R2 | Red band | DoE Lesson 2 (adapted)

Duration: 25-30 min	**Outcomes: R2**	Syllabus: MA3-RN-02, MAO-WM-01 | RN-A: express and interpret decimals to thousandths; place value to 3 dp; trailing zeros

Linked to: DoE Lesson 2 (adapted) -- Lesson 2 -- Representing thousandths | Key resources: Place-value houses, decimal place-value chart, hundredths/thousandths grids

Learning intention	Success criteria
Students are learning to: express thousandths as decimals, interpret decimals to 3 places and name the place value of each digit	Students can: * I can express a number of thousandths as a decimal (e.g. 7 thousandths = 0.007). * I can name the place value of each digit in a number like 2.375. * I know a zero at the end of a decimal does not change its value (0.5 = 0.50).

Mini masterclass structure (25-30 min)

Activate (3 min): Extend the place-value chart past hundredths to thousandths. Ask how small a thousandth is.

Model (10 min): Express 124 thousandths as 0.124, name the place value of each digit in 2.375, and show 0.5 = 0.50 = 0.500.

Guided practice (10 min): Students express thousandths as decimals, name place values and interpret trailing zeros using place-value houses.

Check (4 min): Exit question: write 53 thousandths as a decimal; in 4.609 what is the value of the 6 and the 9?

Key vocabulary	Differentiation
decimal, tenths, hundredths, thousandths, place value, three decimal places, trailing zero, digit	**Support:** Use Resource-style place-value houses; start at hundredths then extend to thousandths. **Extension:** Compare and order decimals to 3 places and place them on a number line.

Assessment	Teaching notes (annotate here)
Students express and interpret decimals to thousandths — links to R2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 3 Understanding percentages [HANDS-ON]

Y1 | Yellow band | DoE Lesson 3 (adapted)

Duration: 30 min	**Outcomes: Y1**	Syllabus: MA3-RN-03, MAO-WM-01 | RN-B: % means percent, 100% is the whole; common percentages as fractions and decimals

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Representing percentages | Key resources: 100-grids, percentage bars, fraction/decimal/percentage matching cards

Physical materials needed: 100-grids, percentage bars, fraction/decimal/percentage cards

Learning intention	Success criteria
Students are learning to: understand that % means percent and represent common percentages as fractions and decimals	Students can: * I know % means 'out of 100' and 100% is the whole amount. * I can shade a percentage on a 100-grid. * I can write common percentages (e.g. 25%, 50%, 75%) as fractions and decimals.

Mini masterclass structure (30 min)

Activate (4 min): Shade 50% of a 100-grid. Ask what 100% would look like and what % means.

Model (10 min): Show 25% = 25/100 = 1/4 = 0.25 on a 100-grid and a percentage bar. Repeat for 50% and 10%.

Guided practice (12 min): Students match common percentages with their fraction and decimal forms and shade them on grids.

Check (4 min): Exit question: write 75% as a fraction and a decimal; what percentage is the whole?

Key vocabulary	Differentiation
percent, percentage, per cent, whole, hundred, fraction, decimal, 100-grid	**Support:** Use a pre-divided 100-grid; provide a fraction-decimal-percentage matching strip for benchmarks. **Extension:** Represent percentages of lengths and quantities, and order a mixed set of fractions, decimals and percentages.

Assessment	Teaching notes (annotate here)
Students represent common percentages as fractions and decimals — links to Y1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 4 Finding 10% of a quantity

Y2 | Yellow band | DoE Lesson 3 (adapted)

Duration: 25 min	**Outcomes: Y2**	Syllabus: MA3-RN-03, MAO-WM-01 | RN-B: 10% is one-tenth of 100%; find 10% of a quantity

Linked to: DoE Lesson 3 (adapted) -- Lesson 3 -- Representing percentages (finding 10%) | Key resources: 100-grids, place-value chart

Learning intention	Success criteria
Students are learning to: use the fact that 10% is one-tenth of 100% to find 10% of a quantity	Students can: * I know 10% is one-tenth of the whole. * I can find 10% of a quantity by dividing by 10. * I can use 10% to estimate other percentages (e.g. 20% = double 10%).

Mini masterclass structure (25 min)

Activate (3 min): Shade 10% of a 100-grid. Ask what fraction that is (one-tenth).

Model (10 min): Find 10% of 200 by dividing by 10 (20). Find 10% of $45. Then find 20% as double 10%.

Guided practice (9 min): Students find 10% of several quantities by dividing by 10, and use it to find 20% and 5%.

Check (3 min): Exit question: find 10% of 350 and use it to find 20% of 350.

Key vocabulary	Differentiation
10 per cent, one-tenth, divide by 10, quantity, whole, estimate	**Support:** Use a place-value shift (divide by 10) with concrete amounts of money. **Extension:** Use 10% as a building block to find 30%, 15% or 5% of a quantity.

Assessment	Teaching notes (annotate here)
Students find 10% of a quantity using the one-tenth relationship — links to Y2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 5 Percentage of a quantity and percentage change (Stage 4)

G1 | Green band | Original mini lesson

Duration: 30-35 min	**Outcomes: G1**	Syllabus: MA4-FRC-C-01, MAO-WM-01 | Stage 4 FRC-A: percentage of a quantity; percentage increase and decrease

Learning intention	Success criteria
Students are learning to: find a percentage of a quantity and calculate percentage increases and decreases	Students can: * I can find a percentage of a quantity (e.g. 15% of 80). * I can express one quantity as a percentage of another. * I can calculate a simple percentage increase or decrease.

Mini masterclass structure (30-35 min)

Activate (4 min): Recall finding 10% by dividing by 10 (Mini Lesson 4). Ask how to find 15% using 10% and 5%.

Model (12 min): Find 15% of 80 (10% = 8, 5% = 4, so 15% = 12). Express 12 out of 80 as a percentage. Calculate a 20% increase on $50 and a 25% discount on $40.

Guided practice (12 min): Students find percentages of quantities, express one quantity as a percentage of another, and calculate increases/decreases in shopping contexts (AUD).

Check (4 min): Exit question: find 25% of 60; a $200 item is reduced by 10% — what is the new price?

Key vocabulary	Differentiation
percentage, of a quantity, percentage increase, percentage decrease, discount, original amount	**Support:** Build percentages from 10% and 5%; provide a worked frame for increase/decrease. **Extension:** Calculate the original price before a discount; solve multi-step percentage word problems including GST (10%).

Assessment	Teaching notes (annotate here)
Students find percentages of quantities and calculate percentage change — links to G1 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 6 Number patterns from multiples [HANDS-ON]

Y3 | Yellow band | DoE Lesson 4 (adapted)

Duration: 30 min	**Outcomes: Y3**	Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: create a table of values from a geometric pattern; describe as multiplication

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Multiplicative patterns | Key resources: Pattern tiles or counters, grid paper, table-of-values templates

Physical materials needed: pattern tiles or counters, grid paper, table-of-values templates

Learning intention	Success criteria
Students are learning to: build a table of values from a geometric pattern and describe the pattern as multiplication	Students can: * I can build the next terms of a geometric pattern that grows by multiplying. * I can record the pattern in a table of values. * I can describe the pattern in words as multiplication (e.g. each term is 3 times the position).

Mini masterclass structure (30 min)

Activate (4 min): Show a tile pattern where each step is a multiple (3, 6, 9...). Ask how the pattern grows.

Model (10 min): Build a pattern with tiles, record term number and tile count in a table, and describe it as 'tiles = 3 x term', emphasising multiplication not addition.

Guided practice (12 min): Students build/extend a geometric pattern, complete a table of values and describe the pattern in multiplication terms.

Check (4 min): Exit question: complete the table for a pattern of multiples of 4 and describe it in words.

Key vocabulary	Differentiation
pattern, multiple, term, position, table of values, multiplication, geometric pattern	**Support:** Provide the pattern built for the first 3 terms and a part-filled table. **Extension:** Compare two patterns with different multiples and describe how their tables differ.

Assessment	Teaching notes (annotate here)
Students build a table of values and describe a multiplicative pattern — links to Y3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 7 Finding the rule of a table

Y4 | Yellow band | DoE Lesson 4 (adapted)

Duration: 25-30 min	**Outcomes: Y4**	Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: determine a rule relating the two rows of a table (algebraic reasoning)

Linked to: DoE Lesson 4 (adapted) -- Lesson 4 -- Multiplicative patterns (finding the rule) | Key resources: Table-of-values templates, function-machine cards

Learning intention	Success criteria
Students are learning to: determine a rule describing the relationship between the bottom number and the top number in a table	Students can: * I can look across a table to see how the top number relates to the bottom number. * I can write the rule as a multiplication (e.g. top = bottom x 5). * I can use the rule to find a missing value.

Mini masterclass structure (25-30 min)

Activate (4 min): Show a table (1->5, 2->10, 3->15). Ask what you do to the bottom number to get the top.

Model (10 min): Establish the rule 'top = bottom x 5' and test it on each column. Use it to find the value for bottom = 8.

Guided practice (10 min): Students find the rule for several tables and use it to complete missing values.

Check (3 min): Exit question: find the rule for 2->8, 3->12, 5->20 and use it to find the value for 7.

Key vocabulary	Differentiation
rule, relationship, table of values, multiply, times, term, function machine	**Support:** Use a function-machine card showing 'x ?'; restrict to single-step multiplication rules. **Extension:** Find rules of the form 'multiply then add' and check them against every column.

Assessment	Teaching notes (annotate here)
Students determine the rule of a table and use it to find missing values — links to Y4 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 8 Multiplying 2- and 3-digit numbers [HANDS-ON]

R3 | Red band | DoE Lesson 5 (adapted)

Duration: 30-35 min	**Outcomes: R3**	Syllabus: MA3-MR-01, MAO-WM-01 | MR-A: factorise; area model; multiplication algorithm; word problems

Linked to: DoE Lesson 5 (adapted) -- Lesson 5 -- Multiplication word problems | Key resources: Grid paper for the area model, place-value chart

Physical materials needed: grid paper (area model), place-value chart

Learning intention	Success criteria
Students are learning to: multiply 2- and 3-digit numbers by 2-digit numbers using the area model and the algorithm	Students can: * I can factorise a number to make a multiplication easier. * I can use the area model to multiply a 2-digit number by a 2-digit number. * I can use the multiplication algorithm and solve word problems.

Mini masterclass structure (30-35 min)

Activate (4 min): Factorise to multiply 24 x 5 as 24 x 5 (or 12 x 10). Ask how splitting numbers helps.

Model (12 min): Model 34 x 26 with the area model (30+4)(20+6), add the partial products, then show the same with the algorithm. Set it in a word-problem context.

Guided practice (14 min): Students multiply 2- and 3-digit numbers using the area model and the algorithm, then solve a multiplication word problem.

Check (4 min): Exit question: use the area model or algorithm to find 47 x 23 and write a word problem it could answer.

Key vocabulary	Differentiation
factorise, area model, partial product, algorithm, multiply, 2-digit, 3-digit, word problem	**Support:** Provide a labelled area-model grid; start with 2-digit by 1-digit before 2-digit by 2-digit. **Extension:** Multiply 3-digit by 2-digit numbers and check using a different strategy (estimation or factorising).

Assessment	Teaching notes (annotate here)
Students multiply using the area model and algorithm and solve word problems — links to R3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 9 Dividing and interpreting remainders

R4 | Red band | DoE Lesson 6 (adapted)

Duration: 30 min	**Outcomes: R4**	Syllabus: MA3-MR-01, MAO-WM-01 | MR-A: division word problems; interpret remainders; digital technologies

Linked to: DoE Lesson 6 (adapted) -- Lesson 6 -- Division word problems | Key resources: Place-value chart, calculators (digital technologies)

Learning intention	Success criteria
Students are learning to: divide a 3+ digit number by a one-digit divisor and interpret the remainder in context	Students can: * I can divide a 3-digit (or larger) number by a one-digit divisor. * I can record my strategy and check with multiplication. * I can interpret a remainder in a word problem (round up, round down or as a fraction).

Mini masterclass structure (30 min)

Activate (4 min): Share 154 items between 6 groups. Ask what to do with the leftover (remainder).

Model (10 min): Model 154 ÷ 6 = 25 r 4, then interpret the remainder in two word problems (e.g. how many full buses vs how much is left over).

Guided practice (12 min): Students solve division word problems, interpret the remainder appropriately, and check with a calculator (digital technologies).

Check (4 min): Exit question: 250 cupcakes in boxes of 8 — how many full boxes, and how many left over?

Key vocabulary	Differentiation
divide, divisor, quotient, remainder, interpret, round up, round down, word problem	**Support:** Provide the division partly worked; use concrete sharing for the remainder. **Extension:** Divide by 2-digit divisors using digital technologies and interpret remainders as fractions or decimals.

Assessment	Teaching notes (annotate here)
Students divide and interpret remainders in context — links to R4 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 10 Order of operations and grouping symbols (Stage 4)

G2 | Green band | Original mini lesson

Duration: 30 min	**Outcomes: G2**	Syllabus: MA4-INT-C-01, MAO-WM-01 | Stage 4 INT-A: apply the 4 operations; grouping symbols; order of operations

Learning intention	**Success criteria**
Students are learning to: apply the order of operations, including grouping symbols, to evaluate expressions	Students can: * I know the order of operations (brackets, then x and ÷, then + and -). * I can evaluate an expression with more than one operation. * I can use grouping symbols (brackets) to change the order.

Mini masterclass structure (30 min)

Activate (4 min): Show 3 + 4 x 5. Ask whether it is 35 or 23, and why order matters.

Model (11 min): Establish the order of operations. Evaluate 3 + 4 x 5 = 23, then (3 + 4) x 5 = 35, then a 3-operation expression with brackets.

Guided practice (11 min): Students evaluate expressions with the four operations and grouping symbols, and insert brackets to make a given target.

Check (4 min): Exit question: evaluate 20 - (3 + 2) x 2; where would you put brackets in 6 + 4 x 2 to make 20?

Key vocabulary	**Differentiation**
order of operations, grouping symbols, brackets, evaluate, expression, operation	**Support:** Provide the order-of-operations reminder; do one operation type at a time before mixing. **Extension:** Insert brackets to reach a target value and justify; evaluate expressions including division and remainders.

Assessment	**Teaching notes (annotate here)**
Students apply the order of operations with grouping symbols — links to G2 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 11 Multiplying and dividing decimals by powers of 10

Y5 | Yellow band | DoE Lesson 7 (adapted)

Duration: 25-30 min	**Outcomes: Y5**	Syllabus: MA3-MR-01, MAO-WM-01 | MR-B: compare relative place value to multiply/divide a decimal by powers of 10

Linked to: DoE Lesson 7 (adapted) -- Lesson 7 -- Multiplying and dividing decimals by powers of 10 | Key resources: Place-value chart / slider, decimal cards

Learning intention	Success criteria
Students are learning to: multiply and divide decimals by powers of 10 by shifting the digits' place value	Students can: * I know multiplying by 10 makes each digit 10 times larger (shifts left). * I can multiply a decimal by 10, 100 or 1000. * I can divide a decimal by 10, 100 or 1000.

Mini masterclass structure (25-30 min)

Activate (4 min): Show 3.5 x 10 on a place-value chart. Ask what happens to each digit's place value.

Model (10 min): Multiply 3.5 x 10 = 35 and 0.24 x 100 = 24 by shifting digits; then divide 47 ÷ 10 = 4.7 and 6 ÷ 100 = 0.06. Stress it is a place-value shift, not 'adding zeros'.

Guided practice (10 min): Students multiply and divide decimals by 10, 100 and 1000 using a place-value chart/slider.

Check (4 min): Exit question: 2.4 x 100 = ? and 58 ÷ 1000 = ?

Key vocabulary	Differentiation
power of 10, multiply, divide, place value, shift, decimal, ten/hundred/thousand times	**Support:** Use a place-value slider to physically shift digits; start with x10 and ÷10. **Extension:** Multiply and divide by 10 000 and explain the place-value shift; combine with measurement conversions.

Assessment	Teaching notes (annotate here)
Students multiply and divide decimals by powers of 10 — links to Y5 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 12 Missing numbers and inverse operations

Y6 | Yellow band | DoE Lesson 8 (adapted)

Duration: 25-30 min	**Outcomes: Y6**	Syllabus: MA3-MR-02, MAO-WM-01 | MR-B: complete multi-operation number sentences; use inverse operations

Linked to: DoE Lesson 8 (adapted) -- Lesson 8 -- Calculating missing numbers | Key resources: Number-sentence cards, balance model

Learning intention	Success criteria
Students are learning to: complete number sentences with missing numbers using inverse operations	Students can: * I can find a missing number in a number sentence. * I can use the inverse operation to undo an operation. * I can complete a sentence that has more than one operation.

Mini masterclass structure (25-30 min)

Activate (4 min): Show 7 x __ = 56. Ask how dividing (the inverse) finds the missing number.

Model (10 min): Solve 7 x __ = 56 using ÷, then a 2-operation sentence (__ x 3 + 4 = 19) by undoing operations in reverse order.

Guided practice (10 min): Students complete one- and two-operation number sentences using inverse operations, checking by substitution.

Check (4 min): Exit question: find the missing number in __ ÷ 4 = 12 and in 5 x __ - 3 = 32.

Key vocabulary	Differentiation
number sentence, missing number, inverse operation, undo, balance, equivalent	**Support:** Use a balance model; restrict to single-operation sentences first. **Extension:** Write a two-operation missing-number sentence for a partner and check using inverse operations.

Assessment	Teaching notes (annotate here)
Students find missing numbers using inverse operations — links to Y6 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

Mini Lesson 13 Solving simple equations (Stage 4)

G3 | Green band | Original mini lesson

Duration: 30-35 min	**Outcomes: G3**	Syllabus: MA4-EQU-C-01, MAO-WM-01 | Stage 4 EQU-A: pronumerals; equations; solve up to 2-step linear equations

Learning intention	Success criteria
Students are learning to: write a number sentence as an equation using a pronumeral and solve simple linear equations	Students can: * I can use a letter (pronumeral) to stand for an unknown. * I can solve a one-step equation using inverse operations. * I can solve a two-step equation (e.g. 3x + 4 = 19).

Mini masterclass structure (30-35 min)

Activate (4 min): Rewrite '__ x 3 + 4 = 19' (Mini Lesson 12) as 3x + 4 = 19. Introduce x as a pronumeral.

Model (12 min): Solve x + 7 = 12 (one step), then 3x + 4 = 19 by undoing in reverse (subtract 4, divide by 3), and verify by substitution. Model one word problem as an equation.

Guided practice (12 min): Students solve one- and two-step equations using inverse operations and verify each solution by substitution.

Check (4 min): Exit question: solve 2x + 5 = 17 and check your answer.

Key vocabulary	Differentiation
equation, pronumeral, unknown, solve, inverse operation, substitute, verify, two-step	**Support:** Connect each equation back to the missing-number sentence; provide the inverse-operation steps. **Extension:** Solve equations with pronumerals on both sides; model and solve a word problem with a 2-step equation.

Assessment	Teaching notes (annotate here)
Students solve simple linear equations using inverse operations — links to G3 in the SOLO Show task.	Taught as described / adapted / replaced -- note here:

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Unit 31 -- Resource Appendix

Every link verified. Videos open in a browser; worksheets are print-ready PDFs (see the download checklist at the end).

Type	Resource	Link
R1 -- Partition, regroup and rename whole numbers to 1 billion.		
Video	Whole Number Expanded Form, Word Form and Standard Form | Math with Mr. J	https://www.youtube.com/watch?v=NOx6jlVSSpU
Video	Partitioning Numbers in Standard and Non-Standard Forms | Learn and Play Online!	https://www.youtube.com/watch?v=vZcR6VY9pDg
Worksheet (print)	Corbettmaths — Place Value (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
R2 -- Express and interpret decimals to thousandths (place value to 3 dp; trailing zeros).		
Video	Place Value to the Thousandths | Miacademy	https://www.youtube.com/watch?v=BsZJGMRWEK0
Video	Finding the Value of the Underlined Digit (Decimal Place Value) | Math with Mr. J	https://www.youtube.com/watch?v=2xGzQXn3WUQ
Video	Expanded Form with Decimals | Math with Mr. J	https://www.youtube.com/watch?v=GX8o-S6Vxig
R3 -- Multiply 2- and 3-digit numbers by 2-digit numbers (area model, algorithm, word problems).		
Video	2-Digit by 2-Digit Multiplication | Math with Mr. J	https://www.youtube.com/watch?v=PZjIT9CH6bM
Video	3-Digit by 2-Digit Multiplication | Math with Mr. J	https://www.youtube.com/watch?v=dNSrfoYLQZU
Video	Box Method Multiplication (2-Digits x 2-Digits) | Math with Mr. J	https://www.youtube.com/watch?v=n3q3XzzIGSY
Worksheet (print)	Corbettmaths — Multiplication (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf
R4 -- Divide a 3+ digit number by a one-digit divisor, interpreting remainders.		
Video	How to do Long Division (1-Digit Divisors) | Math with Mr. J	https://www.youtube.com/watch?v=up_xKZ6GeUg
Video	Box Method for Long Division (1-Digit Divisors) | Math with Mr. J	https://www.youtube.com/watch?v=8JYfVv94hww
Video	Interpreting Remainders in Division Word Problems | LaFountaine of Knowledge	https://www.youtube.com/watch?v=0dPQEKjTFjg
Worksheet (print)	Corbettmaths — Division (PDF)	https://corbettmaths.com/wp-content/uploads/2018/01/division-pdf.pdf
Y1 -- Percentages: % and 100% as the whole; common percentages as fractions and decimals.		
Video	Converting Between Fractions, Decimals and Percents (Mini Course) | Math with Mr. J	https://www.youtube.com/watch?v=2vjbNZaFz3c
Video	Fractions to Percents | Math with Mr. J	https://www.youtube.com/watch?v=XXf4wHb2lYc
Video	How to Convert Decimals to Percents | Math with Mr. J	https://www.youtube.com/watch?v=84WDtZwrGhA
Website	Maths is Fun — Percentages (out of 100, as fractions and decimals)	https://www.mathsisfun.com/percentage.html
Worksheet (print)	Corbettmaths — Fractions, Decimals & Percentages (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf
Y2 -- Find 10% of a quantity (10% is one-tenth of 100%).		
Video	Using 10% to Work Out Percentages | Maths with Mrs B	https://www.youtube.com/watch?v=EM8X3SzReTs
Worksheet (print)	Corbettmaths — Percentage of an Amount (PDF; start with the 10% questions)	https://corbettmaths.com/wp-content/uploads/2018/11/Percentages-of-amounts-234-pdf.pdf
Y3 -- Build a table of values from a geometric pattern and describe it as multiplication.		
Video	Input Output Tables: Find Patterns or Rules | Easy Math with Mrs. Easley	https://www.youtube.com/watch?v=q0n7FMe1CQk
Y4 -- Determine the rule relating the two rows of a table (algebraic reasoning).		
Video	Input Output Tables: Find Patterns or Rules | Easy Math with Mrs. Easley	https://www.youtube.com/watch?v=q0n7FMe1CQk
Video	Input & Output Tables: Rules & Missing Values | STEAMspirations	https://www.youtube.com/watch?v=Z7UPvsKz4Kw
Video	Input Output Tables Song: Finding the Rule of a Function | NUMBEROCK	https://www.youtube.com/watch?v=d1COIcRoSgo

Type	Resource	Link
Y5 -- Multiply and divide decimals by powers of 10.		
Video	Multiplying and Dividing by Powers of 10 | Math with Mr. J	https://www.youtube.com/watch?v=HH-yrNS80Cg
Video	Multiplying and Dividing Decimals by 10, 100 and 1000 | Twinkl	https://www.youtube.com/watch?v=TIlAg3Aa5rA
Y6 -- Complete number sentences with missing numbers using inverse operations.		
Video	How Do You Solve Inverse Operations? (KS2) | Twinkl	https://www.youtube.com/watch?v=enw_kJRSt4A
Video	Solving Find the Missing Number Problems | Maths Junction	https://www.youtube.com/watch?v=rdzDsbKXA3Q
Website	Maths is Fun — Introduction to Algebra (interactive)	https://www.mathsisfun.com/algebra/introduction.html
G1 -- Find a percentage of a quantity; calculate percentage increase and decrease.		
Video	How to Find a Percent of a Number | Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Video	Calculating Percent Increase | Math with Mr. J	https://www.youtube.com/watch?v=t4mWCKu44lU
Video	Calculating Percent Decrease | Math with Mr. J	https://www.youtube.com/watch?v=kS_YxXf6gNA
Video	Finding What Percent One Number Is of Another | Math with Mr. J	https://www.youtube.com/watch?v=Y3g3mourmwQ
Worksheet (print)	Corbettmaths — Percentage of an Amount (PDF)	https://corbettmaths.com/wp-content/uploads/2018/11/Percentages-of-amounts-234-pdf.pdf
G2 -- Apply the order of operations with grouping symbols to the four operations.		
Video	Order of Operations (No Parentheses or Exponents) | Math with Mr. J	https://www.youtube.com/watch?v=BCg1nz5k0HY
Video	Order of Operations (Primary) | Corbettmaths	https://www.youtube.com/watch?v=IKQgvJda7TE
Worksheet (print)	Corbettmaths — Order of Operations (PDF; skip questions with powers like 3^2)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
G3 -- Solve simple (up to 2-step) linear equations using pronumerals and inverse operations.		
Video	An Intro to Solving Two-Step Equations | Math with Mr. J	https://www.youtube.com/watch?v=0ackz7dJSYY
Video	How to Solve Two-Step Equations | Math with Mr. J	https://www.youtube.com/watch?v=7hiNNvqJNos
Video	How to Solve Equations (One-Step & Two-Step) | Math with Mr. J	https://www.youtube.com/watch?v=9eRF7pJCOUQ
Worksheet (print)	Corbettmaths — Equations (PDF; questions 1–14 are one- and two-step)	https://corbettmaths.com/wp-content/uploads/2013/02/equations-pdf.pdf
Beyond SOLO -- Stage 4 extension		
Video	How to Find a Percent of a Number | Math with Mr. J	https://www.youtube.com/watch?v=5CAncQ4TZD4
Video	Calculating Percent Increase | Math with Mr. J	https://www.youtube.com/watch?v=t4mWCKu44lU
Video	Order of Operations (Primary) | Corbettmaths	https://www.youtube.com/watch?v=IKQgvJda7TE
Video	An Intro to Solving Two-Step Equations | Math with Mr. J	https://www.youtube.com/watch?v=0ackz7dJSYY
Worksheet (print)	Corbettmaths — Order of Operations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
Worksheet (print)	Corbettmaths — Equations (PDF)	https://corbettmaths.com/wp-content/uploads/2013/02/equations-pdf.pdf
Website	Maths is Fun — Percentage Change (interactive)	https://www.mathsisfun.com/numbers/percentage-change.html
Website	Maths is Fun — Order of Operations (BODMAS)	https://www.mathsisfun.com/operation-order-bodmas.html
Website	Maths is Fun — Solving Equations (interactive)	https://www.mathsisfun.com/algebra/equations-solving.html
Website	Maths is Fun — Percentages (interactive)	https://www.mathsisfun.com/percentage.html

Worksheet download checklist

Download and print these PDFs before the unit (or save to your class drive). Tick each as you go.

- ☐ Corbettmaths — Place Value (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/place-value-pdf.pdf
- ☐ Corbettmaths — Multiplication (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/multiplication-pdf.pdf

- [ ] Corbettmaths — Division (PDF) -- https://corbettmaths.com/wp-content/uploads/2018/01/division-pdf.pdf
- [ ] Corbettmaths — Fractions, Decimals & Percentages (PDF) -- https://corbettmaths.com/wp-content/uploads/2013/02/fractions-decimals-percentages-pdf.pdf
- [ ] Corbettmaths — Percentage of an Amount (PDF; start with the 10% questions) -- https://corbettmaths.com/wp-content/uploads/2018/11/Percentages-of-amounts-234-pdf.pdf
- [ ] Corbettmaths — Order of Operations (PDF; skip questions with powers like 3^2) -- https://corbettmaths.com/wp-content/uploads/2013/02/order-of-operations-pdf.pdf
- [ ] Corbettmaths — Equations (PDF; questions 1–14 are one- and two-step) -- https://corbettmaths.com/wp-content/uploads/2013/02/equations-pdf.pdf